Answer Key & Teacher Notes

Session 1 — Factors and Greatest Common Factor · Session 2 — Multiples and Least Common Multiple

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Factors and Greatest Common Factor

TALK PROMPTS

- Why does the greatest common factor give the greatest number of equal bags?
- A smaller shared factor also works. Why does the problem ask for the greatest one?

PARTNER TASK

Factor Pairs and Trees. One partner lists all factor pairs of 45; the other builds a factor tree for 42. Swap and check, then find the GCF of 28 and 40 together (Garden Rows: 4 rows).

WATCH FOR

- Choosing a factor of only one number (GCF of 9 and 30 is 9) without checking that it divides both.
- Giving a multiple instead of a factor (confusing GCF with LCM).
- Stopping at a common factor that is not the greatest, or skipping 1 and the number itself when listing factors.

SESSION 2 · TEACHER NOTES

Multiples and Least Common Multiple

TALK PROMPTS

- Why does this question need the least common multiple instead of any common multiple?
- Your partner picks 108 because 9 × 12 = 108. That is a shared multiple. Why is it the wrong answer here?

PARTNER TASK

Skip-Count Lists. One partner lists the first five multiples of 7; the other lists the first five multiples of 4. Find the LCM together and check whether it equals the product. Repeat with 6 and 8 and compare.

WATCH FOR

- Using the product as the LCM (9 × 12 = 108 instead of 36).
- Choosing a later shared multiple instead of the first one on both lists.
- Mixing up GCF and LCM: answering a "next time together" question with a factor.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.B.4	1	1	1	2	2	2	2	3	2	1	1
S1 · Form B	6.NS.B.4	1	1	1	2	2	2	2	3	2	1	1
S2 · Form A	6.NS.B.4	1	1	2	1	2	2	2	3	2	1	1
S2 · Form B	6.NS.B.4	1	1	2	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

factor: A whole number that divides another number with **no** remainder.

common factor: A factor that two numbers **share**.

greatest common factor (GCF): The **greatest** factor that two numbers share.

prime factorization: A number written as a product of **prime** numbers.

- List the factors of each number, then circle the factors that appear in **both** lists.
- The GCF tells the greatest number of **identical** groups you can make with nothing left over.

Example: Factors of 20: 1, 2, 4, 5, **10**, 20. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Shared: 1, 2, 5, 10. The GCF is **10**, so she can make 10 kits. Answer: Each kit has 20 ÷ 10 = **2** pencils and 30 ÷ 10 = **3** markers.

YOU TRY

1. a. 6 b. 9

2. 8 stations (3 balls and 4 cones each)

EXIT TICKET

1. a. 9 b. 10

2. 4 baskets (4 apples and 7 oranges each)

3. **C** (12)

PRACTICE FORM A

1. a. 8 b. 6 c. 7 d. 5

2. a. 1 × 45, 3 × 15, 5 × 9 b. 1, 3, 5, 9, 15, 45

3. a. 2 × 3 × 7 b. Yes, 42 ÷ 7 = 6

4. a. 9 bags b. 2 erasers and 3 stickers

5. No; 3 is not a factor of 25. GCF = 5

6. No; 2 is not a factor of 49. GCF = 7

7. a. 4 rows b. 7 carrot seeds and 10 bean seeds

8. a. 12 bags b. 4 stickers and 5 erasers c. 10 is not a factor of 48 (48 ÷ 10 = 4.8), so stickers would be left over.

9. A common factor must divide BOTH numbers. 9 does not divide 30 (30 ÷ 9 has a remainder). Factors of 9: 1, 3, 9; factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. GCF = 3.

10. **B** (6)

11. **C** (no remainder)

Self-check word **FACTOR**

PRACTICE FORM B

1. a. 6 b. 11 c. 4 d. 7

2. a. 1 × 63, 3 × 21, 7 × 9 b. 1, 3, 7, 9, 21, 63

3. a. 2 × 5 × 7 b. Yes, 70 ÷ 5 = 14

4. a. 5 packs b. 4 chew toys and 7 treats

5. No; 3 is a common factor but not the greatest. GCF = 6

6. No; 2 is not a factor of 55. GCF = 11

7. a. 9 rows b. 4 carrot seeds and 5 bean seeds

8. a. 8 bags b. 5 stickers and 7 erasers c. 10 is not a factor of 56 (56 ÷ 10 = 5.6), so erasers would be left over.

9. Devon only checked 18. 9 does not divide 42 (42 ÷ 9 has a remainder). Shared factors of 18 and 42: 1, 2, 3, 6. GCF = 6.

10. **D** (4)

11. **B** (4)

Self-check word **DIVIDE**

GUIDED NOTES

multiple: A number you land on when you **skip-count** by another number.

common multiple: A number that is a multiple of **both** numbers.

least common multiple (LCM): The **smallest** number above 0 that is a multiple of both.

- List multiples of each number and find the **first** number on both lists.
- The product of two numbers is always a common multiple, but not always the **least** one.

Example: Multiples of 8: 8, 16, 24, 32, **40**. Multiples of 10: 10, 20, 30, 40. The first shared multiple is the LCM: **40**. Answer: Both visit again in **40** days.

YOU TRY

1. a. 12 b. 10

2. 36 of each (3 packs of plates, 4 packs of cups)

EXIT TICKET

1. a. 30 b. 9

2. 20 seconds

3. **B** (When will two buses next arrive together?)

PRACTICE FORM A

1. a. 28 b. 24 c. 18 d. 20

2. a. 15, 30, 45, 60; 20, 40, 60 b. 60 min

3. a. 42 of each b. 3 packs of cups, 2 packs of lids

4. 15 days

5. No; 320 is a common multiple but not the least. LCM = 80

6. a. 150 s b. A: 150 ÷ 25 = 6 stamps; B: 150 ÷ 30 = 5 stamps

7. 30

8. a. 36 b. 7:36 a.m. c. 3 times (7:36, 8:12, 8:48)

9. 30 is a common multiple, but the LCM is the FIRST (least) one on both lists. Multiples of 3: 3, 6, 9, 12, 15; multiples of 5: 5, 10, 15. LCM = 15.

10. **C** (24)

11. **D** (35)

Self-check word **MULTIPLE**

PRACTICE FORM B

1. a. 30 b. 50 c. 24 d. 48

2. a. 12, 24, 36; 18, 36 b. 36 min

3. a. 24 of each b. 4 packs of juice, 3 sleeves of straws

4. 28 days

5. No; 432 is a common multiple but not the least. LCM = 72

6. a. 140 s b. A: 140 ÷ 20 = 7 stamps; B: 140 ÷ 35 = 4 stamps

7. 12

8. a. 40 b. 8:40 a.m. c. 3 times (8:40, 9:20, 10:00)

9. 48 is a common multiple, but the LCM is the FIRST (least) one on both lists. Multiples of 6: 6, 12, 18, 24; multiples of 8: 8, 16, 24. LCM = 24.

10. **C** (15)

11. **D** (36)

Self-check word **QUOTIENT**

SHEET 1 · FACTORS AND GREATEST COMMON FACTOR

1. a. 1, 2, 3, 5, 6, 10, 15, 30 b. 1, 2, 4, 7, 14, 28 c. 1, 3, 5, 9, 15, 45 d. 1, 17 e. 1, 2, 3, 4, 6, 8, 12, 16, 24, 48 f. 1, 2, 3, 6, 7, 14, 21, 42

2. a. 4 b. 9 c. 8 d. 1 e. 12 f. 15

3. a. $2^2 \times 3^2$ b. 2×5^2 c. $2^3 \times 3^2$

4. GCF(36, 54) = 18 bags; each bag has 2 granola bars and 3 juice boxes

5. 1 is a factor of every whole number, so 15 and 28 share the factor 1. The GCF is 1.

SHEET 2 · MULTIPLES AND LEAST COMMON MULTIPLE

1. a. 21 b. 20 c. 24 d. 15 e. 36 f. 50

2. LCM(8, 10) = 40; 5 packs of hot dogs and 4 packs of buns

3. LCM(6, 9) = 18 seconds

4. LCM of 4 and 6 (12 minutes)

5. No. LCM(4, 5) = 20 = 4 × 5, but LCM(4, 6) = 12, not 24. The product is a common multiple, but not always the least (when the numbers share a factor greater than 1).

6. **B** (30)

7. **B** (6 and 8)

October 2026 conceptual additions — teacher key

Session 1 · Factors and Greatest Common Factor

GCF 12; 12 kits, each with 2 pencils and 3 markers. Distinguish kit count from items inside a kit.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: GCF 12; 12 kits, each with 2 pencils and 3 markers. Distinguish kit count from items inside a kit.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Multiples and Least Common Multiple

LCM 24; next shared visit in 24 days. Today is day zero and is excluded from next.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: LCM 24; next shared visit in 24 days. Today is day zero and is excluded from next.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.